ICAR Rabi season Agro Advisory for Maharashtra

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Advisories Related to Major Rabi Crops

Chickpea:

- Rainfed chickpea should be sown in first fortnight of October while irrigated chickpea should be sown from last week of October to first week of November. Latest improved desi varieties like Akash, Phule Vikram, Phule Vikrant, JAKI-9218, etc. should be used for high yield. Kabuli varieties like PKV Kabuli-2, Virat, BDNGK-798, etc should be used. Sowing on BBF will be beneficial.
- For desi varieties, 50-75 kg / ha seed rate should be used based on seed size. In case of

Kabuli and bold varieties, a seed rate of 100 kg/ha should be used.
- The seed should be treated with 4 g Trichoderma or 3 g Carbendazim per kg of seed followed by Rhizobium and PSB biofertilizer inoculation @ 250 g / 10 kg of seed.
- A recommended fertilizer dose of 25 kg N, 50 kg P and 25 kg K should be applied as a basal dose at the time of sowing. Applying fertilizers based on soil testing is advisable to identify and fulfil the need of secondary and micronutrients.
- IPM practices should be followed for the control of Helicoverpa by using 5 Pheromone traps per hectare, one spray of 5 % NSKE at flowering stage and ETL based spray of Quinalphos 25 EC 20 ml or Emamectin benzoate 5 SG @ 4 g or Chlorantraniliprole 18.5 SC @ 3 ml per 10 lit of water.
- Two protective irrigations at flowering and pod filling stage should be given wherever available.

Wheat:

- The timely sown irrigated wheat should be sown in the first fortnight of November. The late sown wheat may be sown upto15th December if water is available. The rainfed wheat should be sown during first fortnight of October using residual moisture available in the soil.
- The improved varieties like Tryambak, Godawari, Tapowan, Parbhani-51, PDKV Washim, PDKV Sardar, AKAW-4647, AKAW1071 should be used for sowing. Rust tolerant varieties should be preferred.
- Wheat crop should be irrigated in the critical stages of irrigation i.e., CRI stage (18-20 days), tillering stage (30-35 days), flowering (65-70 days), grain filling stage (80-85 days) and grain maturity stage (95-100 days). If only one irrigation is available, it should be applied at 40-42 DAS, if two irrigations are available, it should be applied at 18-20 and 60-65 DAS and if three irrigations ae available, it should be applied at 18-20, 40-42 and 60-65 DAS for better yield.
- The recommended dose of 100-120 kg N, 50-60 kg P and 50-60 kg K should be applied for timely sown wheat while 80:40:40 kg NPK/ha should be applied for the late sown wheat. Half of the N and full P and K should be applied as a basal dose at the time of sowing while remaining half N should be applied at the time of first irrigation (18-20 DAS). The rainfed wheat can be fertigated with 40 kg N and 20 kg N as a basal dose.
- The crop should be protected from stem borer by spraying Cypermethrin 10 EC @ 11 ml per 10 lit waters if infestation is observed.

Rabi Sorghum:

- Rabi sorghum crop should be sown on residual moisture after harvest of green gram, black gram, or Soybean. Rainfed Rabi sorghum should be sown during 15th September to 1st October while the under irrigated situation the crop may be sown up to 31st October.
- The recommended spacing of 45 cm x 12.5 cm should be followed for higher yield; the seeds should be treated with fungicide and Gaucho insecticide @ 4 g / kg of seed for the control of stem borer.
- A recommended dose of 80:40:40 kg NPK should be applied to Rabi sorghum of which half N and full P and K should be applied as a basal dose while half of the N should be applied

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after 30 DAS.
- Improved varieties like Parbhani Moti, Parbhani Super Moti, Phule Vasudha, Phule Revati, PKV Kranti, etc should be used for higher yield.

Safflower:

- The rainfed safflower crop should be sown on residual moisture during 1st October to 15th October. Irrigated safflower can be sown up to 15th November.
- A recommended spacing of 45 x 20 cm should be followed for better yield. The crop stand should be maintained by thinning of crop after two weeks from sowing for maintaining the desired plant population.
- Improved varieties like PBN-12, PBN-46, PBN-40 (non-spiny), PKV Pink (AKS-311), NARI-6 (non-spiny), NARI NH-1 (nonspiny) should be used.
- The crop should be protected from the attack of aphids by using 5% NSKE or Dimethoate 30 EC 10 ml per 10 lit of water.

Linseed:

- Use improved high yielding and short duration varieties of linseed like Latur-93, NL-97, etc.
- Use recommended spacing of 30 cm x 15 cm and 10 kg seed/ha. Finish the sowing before 15th October.
- A recommended dose of fertiliser 50 kg N and 25 kg P should be applied, out of which 50 % N should be applied at the time of sowing and 25 kg after one month of sowing as a top dressing should be applied.
- Timely plant protection measures should be followed as per recommendations.

Mustard:

- Mustard can be sown in Maharashtra in the first fortnight of November. Improved varieties like Shatabdi (CAN-9), TPM-1, etc should be used with 5 kg seed rate per hectare. The sowing should be done at 45 cm x 20 cm spacing.
- The crop should be protected from powdery mildew by spraying of Carbendazim @ 10 g per 10 lit of water or wettable sulphur @ 20 g per 10 lit of waterand timely irrigation should be provided.

Sugarcane:

- Pre-seasonal sugarcane should be planted during 1st October to 15th November while seasonal sugarcane should be planted during 1st January to 15th February. 25 % more yield is realised from preseasonal sugarcane.
- Use improved varieties like MS-10001, VSI-08005, Co-86032, Co-8014 (Mahalaxmi), CoM-0265, etc for higher yield
- Use 25-30 thousand two budded sets for planting; maintain row to row spacing of 90-100 cm and plant to plant spacing of 22 cm as per recommendations.
- Sugarcane sets should be treated with recommended pesticides, fungicides and biofertilizers like Acetobacter and PSB before planting and recommended dose of fertilizers 250:115:115 kg NPK/ ha for seasonal sugarcane and 300:170:170 kg NPK / ha should be applied. Nitrogen doses should be applied in four equal splits.

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Advisories Related to Major Horticulture Crops

Pomegranate:

- Take pruning for Ambiya Bahar in Pomegranate. Use improved and high yielding varieties like Bhagwa and Super Bhagwa. Give stress of 30 to 60 days according to soil type. Give light irrigation for flowering. Precaution should be taken so that water stress condition may not occur in pomegranate orchard. Remove suckers shoot in pomegranate orchard. Apply FYM @ 40-50 kg per plant. Apply fertilizer dose of 400:200:200 g / plant after irrigation. Nitrogen should be given in two splits. Take a spray of potassium nitrate @ 15 gm per litre of water.

Citrus (Orange and sweet orange):

- Ambiya bahar management is the key operation in orange and sweet orange in Rabi season.
- Stress management should be done in the month of November December as per soil type. 45 to 60 days water stress should be given as per soil type. Light irrigation should be given after completion of stress.
- Take a spray of potassium nitrate @ 15 gm per litre of water. Application of Bordo-paste should be done on tree trunk in the month of January-February to protect the crop from gummosis and dieback.
- Apply fertilizer dose of 400:200:200 g / plant after irrigation. Nitrogen should be given in two splits.

Mango:

- Organic mulching should be done, and the orchard should be irrigated during early in the morning, at evening or during night hours. To protect the mango fruits from the incidence of fruit fly, install "*Rakshak fruit fly trap*" developed by university @ 4 traps per hectare. Fallen fruits should be collected and destroyed to keep orchard clean.

Banana:

- To protect the crop from the severe cold wave, make a smoke in the orchard to increase the temperature in the evening and apply irrigation at night to raise the temperature. Organic mulching should be done in banana orchard. Irrigate banana orchard during early in the morning, at evening or during nighttime.

Grapes:

- Management of October pruning in grape orchard should be done as per availability of water. During rest period, provide only need based irrigation to protect the existing leaves from drying, and contribute towards increasing the reserves of the vines through photosynthetic activity. Care should be taken to reduce/stop the water in case new growth is observed on the shoot. If rainfall exceeds 2.5 mm, no irrigation should be applied especially during rest period.

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- Improved varieties like Thompson seedless, Tas-E-Ganesh, Sonaka, Super Sonaka, Sharad seedless, etc should be used for planting.

Advisories Related to Major Vegetable Crops

- **Brinjal:** The crop should be planted in September-October in Rabi season. Improved high yielding varieties like Arka Harshita, Arka Unnati, Arka Shirish, Arka Anand, Arka Shirish, Kalpataru, Krishna, Pragati, Vaishali, Phule Harit, Anuradha, Aruna, etc should be used for planting. Control measures should be taken for brinjal shoot and fruit borer. If observed, collect, and destroy caterpillar or Spray Neemark 4% or Cypermethrin 25% EC 5 ml or Chlorpyriphos 20 EC 20 ml OR Deltamethrin 1 % or Trizophos 20 ml per 10 litres of water.
- **Tomato:** Triple disease resistant varieties of tomato like Arka Rakshak or Arka Samrat should be used for higher yield. The crop should be planted in the month of August-September for Rabi season.
- **Onion:** Improvedvarieties like Baswant-780, N-2-4-1, Agri Found Light Red, Phule Samarth, Bhima Super, Bhima Red, Bhima Shakti, Bhima Shubhra, etc should be used for Rabi cultivation. Transplanting should be done in the month of October-November. Seed can also be done by tractor drawn BBF to save the labour charges and time for seedling preparation and transplanting. Apply Malic Hydroxide-40 @3000 ppm solution 15 to 20 days prior to harvesting to increase shelf life of onion. Also stop irrigation before three weeks of harvesting.
- **Okra:** To control powdery mildew in okra, three sprays of 0.03% wettable sulphur should be taken at 10-15 days of interval. Improved varieties like Parbhani Kranti, Arka Anamika, Phule Kirti, Utkarsh, PBNOK-1 should be used for higher yield.

Advisories Related to Major Floriculture Crops:

- **Gaillardia:** Planting should be done through seeds or seedlings in well-drained soil by using varieties like Murgudi, Fulzari etc. Apply 50:50:50kg NPK/ha at the time of sowing, remaining 50kg N apply after 30 days after planting.
- **Marigold:** Marigold should be planted at 60 x 45 cm spacing in well-drained soil. Seedlings should be prepared through seeds of varieties like *Local red, Local yellow, Bangalore local, Orange Crush. Pineapple Crush, Pusa Basanti Genda* etc., Apply 50:50:50kg NPK/ha at the time of sowing, remaining 50kg N apply after 30 days after planting.
- **Chrysanthemum:** Chrysanthemum requires well drained soil. Apply 112000 saplings/ha for plantings of variety Sankar-1, Guldasta, Raja, Piwlirevdi, Pandhrirevdi, Zipri, Birbal sahani, etc. Application 75:50:50kg NPK/ha apply at sowing time remaining 75kg N apply after 30 days after planting.
- **Tuberose:** Tuberose should be planted at 30 x 30 cm spacing in well-drained soil. Application of 50:50:50kg NPK/ha should be done at the time of sowing time. Remaining 50kg N should be applied after 30 days of planting as top dressing. Improved and high yielding varieties like Single, Double, Phule Rajani, Shrungar, Rajat, Kolkata Single, Prajwal, etc should be used.
- **Gladiolus:** Gladiolus should be planted at 30 x 30 cm spacing in well-drained soil. Application of 50:50:50 kg NPK/ha should be done at the time of sowing time. Remaining 50 kg N should be applied after 30 days of planting as top dressing. Improved high yielding

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50 kg N should be applied after 30 days of planting as top dressing. Improved high yielding varieties like Apsara, Meera, Sapna, Punam, Phule Ganesh, Phule Prerana, Phule Tejas, Nilrekha, White Prosperity, etc should be used.

- **Jasmine:** Planting of jasmine should be done by cutting in well drained black soil; while planting apply 20:20:20 gm NPK/plant, after two months apply 20 gm N/plant. After one year Apply 100:50:50kg NPK/ha at the time of pruning of plants. Improved varieties like Single, Double, Motiya, Madanban, Gundmalai, Bela, Bangalore, etc should be used for higher yield.
- **Rose:** Plant graft in Light to medium well drained soil at 150 x 150 cm spacing. Apply 20:15:15 gm NPK/plant at the time of pruning. The crop should be protected from diseases like downy mildew and powdery mildew. Spray of copper Oxychloride 75 WP 20 g or Captan 50 % WP 25 g per 10 lit of water if incidence is observed.

Advisories Related to Livestock and Fisheries

- **Dairy Cattle:** Vaccination of cattle to protect them from deadly viral foot and mouth disease should be completed in SeptemberOctober and March-April twice in the year.
- **Goat:** Vaccination of goats to protect them from deadly viral PPR disease (Bulkandi) disease should be completed once in every three years.
- **Poultry:** Vent pecking is hazardous and economic loss-making bad habits of newly developed poultry breeds like Grampriya, Kaveri, Vanaraja and Giriraja. To avoid this loss, poultry farmers should follow debeaking of birds and sufficient vitamins (A, D, E, K) and minerals (Ca, P, Na) should be given to the birds through feed or water regularly.
- **Fodder management:** To avoid the annual green fodder scarcity to domestic animals, high yielding Napier fodder varieties like BNH10, Phule Gunwant and DHN-6 should be cultivated by farmers on 0.04 ha area per animal.
- **Fisheries:** Farm pods created under NHM, and state department of agriculture should be utilised for fish farming where water is available throughout the year. Mixed fish breeds like Rohu, Katla, Mrigal and Cyprinus can be grown in fishponds. Biofloc fish farming should be adopted by small and marginal farmers.

Advisories Related to Allied Enterprises:

- **Sericulture:** Mulberry variety V-1 should be planted for higher leaf yield and quality cocoons. One-acre leaves will be sufficient to rear 200 DFL batch. About 1700 sq. ft. space will be required for 200 DFL. One spray of growth hormone Serimore @ one ampule / 50 DFL should be applied at the time of fifth feeding after forth mode for increasing yield and quality of cocoons. One spray of growth hormone Sampoorna @ one ampule / 50 DFL should be applied at the time of 5-10 % sampling of cocoon formation for uniform growth of silkworms. Cocoons should be harvested after five days of cocoon formation. Cocoon harvester machine should be used for harvesting of cocoons for saving the labour.

Advisories Related to Post Harvest Management and Value Addition

- **Sweet Orange:** Sweet orange fruits should be kept in cold storage at -6 ^{0}C for 90 days to

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- **Sweet Orange:** Sweet orange fruits should be kept in cold storage at 6° C for 90 days to increase its shelf life. Waxing can also be done.
- **Grapes:** The fresh fruits should be converted into resins. The fruits can also be kept in cold storage to increase its shelf life.
- **Aonla:** Aonla fruits should be processed and converted into value added products like Aonla candy, Aonla pulp, Aonla syrup, Aonla powder, Aonla supari, etc.
- **Pomegranate:** Pomegranates should be used to make value added products like *Anardana*, Pomegranate juice, etc.
- **Vegetables:** Tomatoes should be processed to make tomato ketch up, tomato sauce, etc. Green and red chilli should also be processed to make value added products like chilli sauce, red chilli powder, green chilli paste, etc.
- **Cereals and pulses:** Sorghum and wheat grains should be stored at below 14 % moisture for long term storage. Improved grain storage bags should be used for storage of wheat, rice, etc. Pulses crops like pigeon pea, chickpea, green gram, black gram should be processed to convert into dal for higher returns. Innovative packaging should be used for fetching more prices in the market.

Advisories Related to Farm Mechanization and Soil Water Conservation

- Use minimum tillage operations, avoid ploughing for land preparation to protect land from evaporation losses and to conserve the soil moisture.
- Use BBF planter for sowing of chickpea for in situ moisture conservation.
- Chickpea crop should be irrigated through sprinkler irrigation at critical stages like flowering and pod filling for increasing yield wherever irrigation is available.
- Use of reapers / combine harvesters should be preferred for harvesting of crops like wheat, chickpea, safflower, etc.
- Rabi onion should be sown with CRIDA Tractor drawn Onion Seed Planter to save labour expenses and to achieve more work efficiency.

Advisories Related to Farm Women

- Farm women should use cycle hoe for intercultural operations in field crops to reduce the drudgery.
- Use serrated sickle for harvesting of Jowar, Bajra and similar crops which reduces the drudgery of farm women.
- Shed net farmers may use vegetable transplanter to reduce drudgery in transplanting, time, money & labour cost.
- Each farm woman should develop a model nutritional garden in their farm or home premises to enhance nutritional status and nutrition security of farm family.
- Nutri-cereals should be added in cropping systems. Nutri-cereals like finger millet, fox tail millet, little millet, etc. can be grown in the Rabi season.
- Farm women should grow sorghum crop for *hurda* (tender roasted grains) on small area and sale it with attractive packing to earn extra income. Similar practice can be done in case of chickpea.

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Source : ICAR Rabi agro-advisory for farmers

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